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简单习题详解

一、概念

1. 么半群

2. $o(a) = m \in \mathbb{Z}^+$

3. 同态

二、判断

1. 有左右消去律的半群必为群

2. 18阶群有4阶子群

3. 18阶元 x , x^2 和 x^8 为9阶

4. 置换可以既奇又偶.

三、填空

1. $\mathbb{Z}_{20}$, 6的逆元2. $GL_2(\mathbb{R})$, $\begin{pmatrix} 2 & 3 \\ 8 & 6 \end{pmatrix}$ 逆元3. α, β . 求 α^{-1} , $\alpha \circ \beta$ 4. $f: (\mathbb{Z}, +) \rightarrow (\mathbb{C} \setminus \{0\}, \cdot)$ $f(k) = (\sqrt{-1})^k$ 求 f 的像集, $\ker(f)$

四、问答

1. 群的单位元定义 $\checkmark$ (左么元、右逆元 $\Rightarrow$ 是群)2. $x^2 = e$, 则交换群 (2) 4阶群都是交换群

3. 轮换分解, 并求阶

4. Sylow 定理第一部分

5. $Z(G) = \{x \mid \forall y \in G, xy = yx\}$ 中心(1) $Z(G) \triangleleft G$ (2) $|G| = p^k$, p 素数, $k \in \mathbb{Z}^+$, $Z(G) \neq \{e\}$ (3) $|G| = 8$, 则 $|Z(G)| \neq 4$ (4) $G \times G \rightarrow G$. $g \mapsto x = gxg^{-1}$ $Y = \{x \mid x \in G, \forall g \in G, g \mapsto x = gxg^{-1} = x \Leftrightarrow gx = xg\} = Z(G)$ 不动点 $|G| \equiv |Z(G)| \pmod{p}$ $|Z(G)| \equiv 0 \pmod{p}$ $e \in Z(G)$. $|Z(G)| \geq 1$. 故 $p \mid |Z(G)| \rightarrow \neq 1$.

(3) 若 $|Z(G)|=4$. 左乘作用, 考虑中心的陪集.

左乘 $G \times Z^G \rightarrow Z^G$

$$g \mapsto A = gA$$

$$u = G_Z = \{gZ(G) \mid g \in G\}. \text{ 轨道 } = \{Z(G)g\}. \text{ 由于 } gx = xg$$

$$\text{stab}(Z(G)) = \{g \mid g \in G, gZ(G) = Z(G)\}$$

$$|G|=8, |Z(G)|=4, Z(G) \leq G \Leftrightarrow k \in Z(G), |H|=|Z(G)|=4.$$

$$u = \{gZ(G) \mid g \in G\} = G/Z(G) \quad Z(G) \text{ 的陪集全体.}$$

$$u \text{ 中恰有 } Z(G) \text{ 一个子群. } |u| = |G/Z(G)| = \frac{8}{4} = 2. \quad 2 \text{ 个陪集}$$

$$\text{若 } u \text{ 中无 } G \text{ 子群, } |u| = \frac{|G|}{|\text{stab}(u)|} > 1.$$

$\hookrightarrow 2$ 的幂次

$$H = Z(G), u = \{H, gH\}, gH \neq H \quad g \notin H.$$

$$h \in H, hH = Hh = H. \quad hgH = ghH = gH$$

$$g' \notin H, g'H \neq H \quad g'H = gH, g'gH = H.$$

$$(g')^2 H = H, (g')^2 \in H.$$

$$4 \nmid H \nsubseteq, 4 \nmid G-H \nsubseteq, \text{ 且 } g^2 \in H.$$

简单习题详解(接上)

4.22

有限半群 + 消去律 $\Rightarrow$ 群

$$(\mathbb{Z}^+, \cdot) \quad 2 \quad \frac{1}{2} \in \mathbb{Z}$$

$$o(x) = 18, \quad o(x^2) = \frac{18}{2} = 9 \quad o((x^3)^2) = 9 \quad o(x^6) = \frac{18}{\gcd(6, 18)} = 9$$

$$6 \in \mathbb{Z}_{18}, \quad (\mathbb{Z}_{18}, \oplus) \text{ 幺元 } 0. \quad 6 \oplus 14 = 0.$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} (ad-bc)^{-1}$$

$$\mathbb{Z} \rightarrow \mathbb{C} - \{0\}$$

$$f(k) = (\sqrt{-1})^k, \quad \sqrt{-1}, \sqrt{-1}^2 = 1, \sqrt{-1}^3 = -\sqrt{-1}, \sqrt{-1}^4 = 1.$$

$$\text{Im}(f) = \{f(k) \mid k \in \mathbb{Z}\} = \{\pm 1, \pm \sqrt{-1}\}$$

$$\ker(f) = \{k \mid k \in \mathbb{Z}, f(k) = 1\} \quad (\sqrt{-1})^k = 1 \Leftrightarrow 4 \mid k.$$

$$\begin{pmatrix} 1 & 1 \\ 4 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \quad \text{lcm}(4, 4, 2, 2) = 4$$

$(G, *)$ 结合律

$$\exists e \in G, \text{ s.t. } \forall a \in G, \text{ 有 } ea = a$$

$$\forall a \in G, \exists b \in G, \text{ s.t. } ba = e$$

左幺元是幺元, 左逆元是逆元

$$ax = ay \Rightarrow x = y \quad \text{左消去律成立.}$$

$$(\exists b, \text{ s.t. } ba = e. \quad b(ax) = b(ay), (ba)x = (ba)y, ex = ey, x = y)$$

$$\text{左幺元} \quad \forall a \in G, \text{ 要证 } ae = a. \quad \exists b \in G, \text{ s.t. } ba = e.$$

$$ae = a \Leftrightarrow b(ae) = ba = e \Leftrightarrow (ba)e = e \Leftrightarrow ee = e \quad \checkmark$$

$$\text{逆元} \quad \text{设 } a \in G, \text{ 由条件, } \exists b \in G, \text{ s.t. } ba = e$$

$$\text{对 } b, \text{ 由条件, } \exists c \in G, \text{ s.t. } cb = e$$

$$(cb)a = ea = a. \quad \text{又 } c(ba) = ce = c. \quad \therefore a = c$$

$$2. (1) \forall x \in G, \because x^2 = xx = |e \quad x x^{-1} = |e \quad \therefore x = x^{-1}$$

$$\forall a, b, \quad ab = (ab)^{-1} = b^{-1}a^{-1} = ba$$

$$\text{方: } (ab)^2 = abab = |e$$

$$\text{左边乘 } a, \text{ 右边乘 } b. \quad a(abab)b = a|eb = ab$$

$$a^2(ba)b^2 = ba //$$

$$\underline{E} = |e(ba)|e = ba.$$

$$(2) |G| = 4.$$

四阶群在同构下只有两种 $\rightarrow$ 若 $\forall x \in G, x^2 = |e \Rightarrow G$ 交换

$$\text{若 } \exists y \in G, \text{ s.t. } y^2 \neq |e. \quad o(y) \mid 4 \text{ (Lagrange)}, \quad o(y) \in \{1, 2, 4\}$$

$$y^2 \neq |e \Rightarrow o(y) \neq 1, 2$$

$$\langle \{y\} \rangle = \{|e, y, y^2, y^3\} = G. \text{ 循环群交换}$$

$$a^i a^j = a^{j+i} = a^j a^i \quad \therefore G \text{ 交换}$$

$$\text{方: } G \quad |e, x, y \text{ 两两不同.}$$

$$xy, yx$$

$$\text{若 } xy = |e, \text{ 则 } yx = |e, \quad xy = yx$$

$$\text{若 } xy \neq |e. \quad G = \{|e, x, y, xy\} \quad // \text{ } xy \text{ 不可能} = x, y, \text{ 因为消去律, 且 } x, y \neq |e.$$

$$G = \{|e, x, y, yx\}$$

$$xy = yx$$

$$5. (1) \text{ 验证 } Z(G) \triangleleft G$$

乘法, 求逆封闭

$$a, b \in Z(G), \quad \forall x \in G, \quad (ab)x = a(bx) = a(xb) = (ax)b = (xa)b = x(ab)$$

$$ax = xa. \quad a^{-1}(xa) = a^{-1}(ax) = (a^{-1}a)x = x, \quad xa^{-1} = a^{-1}(xa) a^{-1} = a^{-1}x$$

同右乘 a^{-1}

或用正规子集的定义:

左陪集 = 右陪集

$\forall g \in G, \forall a \in Z(G)$ 要证 $ga g^{-1} \in Z(G)$

$$ga g^{-1} = ag g^{-1} = a$$

(2) p 素, $k \in \mathbb{Z}^+$, p^k 阶群的中心 $\neq \{1\}$.

G 共轭作用在 G 上. $g \mapsto x = g x g^{-1}$

$$\{x \mid x \in G, \forall g \in G, g \mapsto x = x\} = \{x \mid x \in G, \forall g \in G, g x g^{-1} = x\}$$

$$= \{x \mid x \in G, \forall g \in G, gx = xg\} = Z(G).$$

$$\therefore |G| \equiv |Z(G)| \pmod{p} \quad // \text{上课讲了}$$

$$\therefore |Z(G)| \neq 1$$

(3) $|G| = 8$, 证 $|Z(G)| \neq 4$.

$H = Z(G)$. 假设 $|H| = 4$

取 $x \in G - H$. $G = H \cup xH$

$$(x \notin H \Rightarrow xH \cap H = \emptyset)$$

下面说明 $x \in Z(G) = H$.

$$\forall h \in H, xh = hx \quad (h \in Z(G)) \quad 4 \uparrow$$

$$xh, (xh)x = x(hx) = x(xh) \Rightarrow x \in Z(G) \quad 4 \uparrow$$

$H \cup xH$ 覆盖 G . 因为 $h: 4 \uparrow, xh: 4 \uparrow, (x \notin H)$

$Z(G) \neq 1, 4$. $Z(G) = 2, 8$ 均可能.

$(R, +, \cdot)$ 是环

(1) $(R, +)$ 交换群, $\exists$ 元 0

(2) $(R, \cdot)$ 半群, $\forall$ 有 $\neq 0$ 元, 记为 1_R

(3) $\forall a, b, c \in R, (a+b)c = ac + bc$

$$a(b+c) = ab + ac$$

若 $\forall a, b \in R, ab = ba$.

则称 $(R, +, \cdot)$ 是交换环.

$(R, +, \cdot) \quad \forall a \in R, a \neq 0 \quad a^{-1} = \frac{1}{a}$ (倒数), $aa^{-1} = a^{-1}a = 1$.

$(M_2(\mathbb{R}), +, \cdot) \quad \exists$ 元 $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ 无逆元

$(F, +, \cdot)$ 是域, 是指:

✓ (1) 是交换环, 有乘法 $\neq 0$ 元 1_F .

(2) $\forall a \in F, a \neq 0, \exists b \in F, \text{ s.t. } ab = 1_F$

或: (1) $(F, +)$ 是交换群, $\exists$ 元 0

(2) $(F, \cdot)$ 是交换 $\neq 0$ 半群, 且 $(F - \{0\}, \cdot)$ 是群

(3) 分配律

数域

eg. $(\mathbb{Q}, +, \cdot)$ $(\mathbb{R}, +, \cdot)$ $(\mathbb{C}, +, \cdot)$

$(\mathbb{Z}, +, \cdot)$ 不是域: $2 \nmid 1 \in \mathbb{Z}$ (除1, -1, 均无逆元)

eg. $K = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ $(K, +, \cdot)$

交换, 结合, 分配律从 $\mathbb{Q}$ 继承.

封闭: $(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2}$

$(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$

$a + b\sqrt{2}$, 不全为0. $\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{(a + b\sqrt{2})(a - b\sqrt{2})} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \frac{a}{a^2 - 2b^2} + \frac{-b}{a^2 - 2b^2}\sqrt{2}$

eg. $\sqrt[3]{2}$ $L = \{a + b\sqrt[3]{2} + c\sqrt[3]{4} \mid a, b, c \in \mathbb{Q}\}$

$\sqrt{2}$ $x^2 - 2$

向 $\mathbb{Q}$ 中加东西产生域 后面会讲

$\sqrt[3]{2}$ $x^3 - 2$

eg. $\{a + b\pi \mid a, b \in \mathbb{Q}\}$ 是子群但不是子环.

$\pi\pi = \pi^2$ 出去了!

$\{a_0 + a_1\pi + a_2\pi^2 + \dots + a_k\pi^k \mid a_i \in \mathbb{Q}, k \in \mathbb{N}\}$ 不是域 (π 不是 $\mathbb{Q}$ 上多项式的根)

$n \in \mathbb{Z}^+$, $n \geq 2$ $(\mathbb{Z}_n = \{0, 1, \dots, n-1\}, \oplus, \otimes)$ 是交换环. $\oplus$ 么元0, $\otimes$ 么元1

$a \oplus b = (a + b) \% n$ $a \otimes b = (ab) \% n$

是域吗? 看n.

$(\mathbb{Z}_6, \oplus, \otimes)$ 不是域. $2, 3 \in \mathbb{Z}_6$ $2 \otimes 3 = (2 \cdot 3) \% 6 = 0$ // 非0元找不到乘法逆

$\geq$ 找不到逆元 $2 \otimes x = 1$, 则 $(2x) \% 6 = 1$. $2x \equiv 1 \pmod{6}$ $2x \equiv 1 \pmod{2}$ X

通过找6的因子证明不是.

$6 \mid (2x-1)$ X

$$\text{若 } 3 \otimes x = 1, (3x) \% 6 = 1. \quad 3x \equiv 1 \pmod{6} \quad 6 \mid (3x-1) \quad 3 \mid (3x-1) \quad \times$$

$(\mathbb{Z}_5, \oplus, \otimes)$ 是域

$$1, 2, 3, 4$$

$$1 \otimes 1 = 1 \quad 2 \otimes 3 = 1 \quad 4 \otimes 4 = 1$$

命题 $(\mathbb{Z}_n, \oplus, \otimes)$ 是域 $\Leftrightarrow n$ 是素数 无 1, n 外其他因子

$\mathbb{Z}_p, p$ 是素数

$\mathbb{R}[x] = \{a_0 + a_1x + \dots + a_nx^n \mid n \in \mathbb{N}, \text{每 } a_i \in \mathbb{R}\}$ 不是域

乘法么元 1 $x+1$ 无逆元. 人为添加 $\frac{1}{x+1}$ (分式域)

$\mathbb{R}(x) = \left\{ \frac{f}{g} \mid f, g \in \mathbb{R}[x], g \neq 0 \right\}$.

$$\frac{f}{g} \cdot \frac{g}{f} = \frac{f \cdot g}{g \cdot f} = 1 \quad (\text{交换, } fg = gf)$$

$(\mathbb{Z}, +, \cdot)$

$$\text{扩 } \mathbb{Z}, \cdot \quad \begin{cases} \frac{b}{a} + \frac{d}{c} = \frac{bc+ad}{ac} \\ \frac{b}{a} \cdot \frac{d}{c} = \frac{bd}{ac} \end{cases}$$